CLASSIFICATION

country_name

Official name of the country/land administrative area during “year” period,
example: USA, Kingdom of England, Britannia

modern_country_name

modern country name equivalent

year

exact year data

land_size

(Core Land)

- Definition: The primary, contiguous landmass where the capital and the vast majority of the population reside.
- The "Greece/Archipelago" Rule: If the core is inherently an archipelago or surrounded by proximate islands that are highly integrated geographically and politically (e.g., Greece, Japan, the UK's home islands), they are counted as one cohesive "Core."
- Function: This number is the ONLY value used to calculate the continent_ratio.

land_size_plus

(Peripheral/Detached Land)

- Definition: The total area in km² of all sovereign territories, exclaves, or overseas dependencies that are physically detached by vast distances, oceans, or other countries, AND are not part of the integrated core.
- Function: This is a supplementary metric. It is not factored into the continent_ratio.
- Examples:
    - USA: land_size = ~7.8 million km² (Lower 48). land_size_plus = ~1.7 million km² (Alaska + Hawaii + Puerto Rico, etc.).
    - Denmark: land_size = ~43,000 km² (Jutland + core islands). land_size_plus = ~2.16 million km² (Greenland + Faroe Islands).
    - France: land_size = ~543,940 km² (Metropolitan France). land_size_plus = ~89,169 km² (French Guiana, New Caledonia, etc.).

continent_size

continent size in km2 (Eurasia divided to Asia 44 261 000 km2 and Europe 10 498 000 km2)

continent_ratio

land/continet ratio

dominant_soil

Definition: The primary soil classification covering the majority of the land area.

- **Chernozems (Black Earth)**
  - Scientific Core: WRB *Chernozems* / USDA *Mollisols*.
  - Geopolitical Impact: The "Master Soil." Highest natural fertility. Supports massive population density and caloric surplus for export. Historically leads to "Breadbasket" states and major power projection.
  - **Examples:** Ukraine, US Great Plains, Southern Russia.
- **Luvisols (Temperate Soils)**
  - Scientific Core: WRB *Luvisols* / USDA *Alfisols*.
  - Geopolitical Impact: High-stability soil. Very productive in temperate climates. It requires more management (plowing/crop rotation) than Black Earth, which historically fostered settled, technologically advancing agrarian societies.
  - Examples: France, Germany, Eastern USA, parts of China.
- **Fluvisols (Riverlands / Alluvial)**
  - Scientific Core: WRB *Fluvisols* / USDA *Fluvaquents*.
  - Geopolitical Impact: Young soils deposited by rivers. Extremely fertile but requires massive coordination (canals, dikes). Historically the engine of Hydraulic Civilizations and highly centralized, absolute political systems.
  - Examples: Nile Valley, Mesopotamia, North China Plain, Po Valley.
- **Aridisols (Desert / Arid Soils)**
  - Scientific Core: USDA *Aridisols* / WRB *Calcisols*.
  - Geopolitical Impact: Water-limited. Agriculture is impossible without large-scale irrigation. Typically leads to low population density, nomadic lifestyles, or highly concentrated "Oasis" urbanism.
  - Examples: Sahara, Arabian Peninsula, Central Asia.
- **Ferralsols (Tropical Soils)**
  - Scientific Core: WRB *Ferralsols* / USDA *Oxisols*.
  - Geopolitical Impact: Highly leached and acidic. Poor for long-term grain agriculture. Historically limited the growth of dense, centralized empires in the tropics and often pushed economies toward plantation-style exports or shifting cultivation.
  - Examples: Amazon Basin, Congo Basin, parts of Southeast Asia.

soil_ratio

Definition: The numerical ratio of area covered by the what_dominant_soil relative to the total land_size (Core Land).

ore_deposit

(Strategic Resource Control)

Definition: Evaluates the inherent geological wealth and resource potential of the state's territory. This value must remain constant from the nation's foundation (or initial data entry) for that specific geographic core. It ignores the technological capacity of the people at the time to extract it. The goal is to measure how the land's fixed baseline potential eventually influences the evolution of the state's political and economic systems.

Classification Scale (0 to 3):

- **0 - Geologically Poor:** The physical territory is naturally devoid of significant, concentrated veins of critical ores, metals, or energy resources. The land inherently lacks heavy industrial potential. *(Example: The core geography of Japan or the Netherlands).*

- **1 - Moderate / Standard Endowment:** The land contains average, varied geological deposits. It has enough natural potential to support independent development, but lacks massive, anomalous concentrations of wealth. *(Example: The core geography of France or Italy).*
- **2 - Resource Rich / High Potential:** The territory sits atop massive, naturally concentrated deposits of globally significant resources (e.g., vast coal seams, heavy iron belts, massive oil reserves). The land is inherently built for heavy extraction and wealth generation. *(Example: The core geography of the United States, Russia, or the DRC).*
- **3 - Geological Jackpot / Global Anomaly:** The territory encompasses an overwhelming, anomalous concentration of a specific critical global resource, representing a massive disproportionate share of the Earth's total geological crust for that material. *(Example: The Arabian Peninsula's oil fields, the Andean silver veins of Potosí).*
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dominant_landscape

Definition: Identifies the primary geomorphological shape of the Core Land. This simplified classification focuses on the "stage" of history—dictating the speed of movement, the feasibility of infrastructure, and the natural defensive potential of the territory.

Scientific Source: Aligned with Global Landform Classification standards (e.g., Sayre et al., 2014) and the geomorphometric mapping principles used by the USGS and IGU (International Geographical Union).

Standardized Classifications:

- **Plains:** Flat or gently undulating land with low relief. Includes alluvial basins and steppes. Geopolitical Impact: High movement speed; vulnerable to invasion but ideal for centralized agricultural states.
- **Hills:** Moderate relief; rolling terrain and dissected uplands. Includes Karst topographies. Geopolitical Impact: Moderate defensive advantage; often leads to localized political fragmentation.
- **Mountains:** High relief; steep slopes and significant peaks. Geopolitical Impact: Natural fortress barriers; historically creates hard borders and isolated cultural identities.
- **Plateaus:** Elevated land with a relatively level surface (Tablelands). Geopolitical Impact: Defensive isolation combined with internal stability for agriculture (e.g., Ethiopia, Spain).
- **Archipelagic:** Highly fragmented landmasses (islands, fjords, or severely broken coastlines). Geopolitical Impact: Necessitates maritime dominance; prevents traditional land-based central authority (e.g., Greece, Norway, Japan).

landscape_ratio

Definition: The numerical ratio of the area covered by the what_dominant_landscape to the total land_size (Core Land).

Interpretation:

- **High (> 0.75):** Geographic monolith; uniform terrain simplifies infrastructure and central control.
- **Low (< 0.40):** Geographic fragmentation; diverse terrain often necessitates federalist or complex governance.

natural_forestation_ratio

Definition: The numerical ratio representing the Actual forest cover of the Core Land as of the specified year. This represents the "texture" of the landscape.
Scientific Source: Aligned with the FAO Global Forest Resources Assessment (FRA)
Calculation: Forestation Ratio=land_size (Core Land)Total Forested Area (km²)

Scientific Logic (The "Jungle" Rule): This metric allows the model to derive complex biomes without additional columns.

- Jungle/Rainforest = what_climate_dominant: Tropical (A) + natural_forestation_ratio: >0,70.
- Taiga/Boreal Forest = what_climate_dominant: Continental (D) + natural_forestation_ratio: >0,60.
- Mixed / Temperate Woodlands = what_climate_dominant: Continental (D) or Temperate (C) + natural_forestation_ratio as r: 0.6 < r > 0.15
- Steppe/Savanna = what_climate_dominant: Arid (B) or Temperate (C) + natural_forestation_ratio: <0,15.

max_degree_north

Measured only from the northernmost and southernmost points of the contiguous Core Land (e.g., USA stops at Washington state/Minnesota, ignoring Alaska).

max_degree_south

Measured only from the northernmost and southernmost points of the contiguous Core Land (e.g., USA stops at Washington state/Minnesota, ignoring Alaska).

climate_dominant

What climate Tier 1 is dominant by percentage (according to Tier 1) - Köppen climate classification
Scientific Source: Köppen-Geiger Climate Classification System (Peel, Finlayson, and McMahon, 2007).

Tier 1		Tier 2		Tier 3	
A	**Tropical**	f m w s	**Rainforest** **Monsoon** **Savanna, dry winter** **Savanna, dry summer**		
B	**Dry**	W S	**Arid desert** **Semi-arid steppe**	h k	**Hot** **Cold**
C	**Temperate**	w f s	**Dry winter)** **No dry season** **Dry summer**	a b c	**Hot summer** **Warm summer** **Cold summer)**
D	**Continental)**	w f s	**Dry winter)** **No dry season** **Dry summer**	a b c d	**Hot summer** **Warm summer** **Cold summer** **Very cold winter**
E	**Polar**	T F	**Tundra** **Ice cap**		

Calculated based only on the weather patterns of the Core Land (e.g., Denmark is Temperate C-tier, completely ignoring Greenland's E-tier Polar climate).

climate_t2_dominant

What climate Tier 2 is dominant by percentage (according to Tier 2) - Köppen climate classification
Scientific Source: Köppen-Geiger Climate Classification System (Peel, Finlayson, and McMahon, 2007).

Tier 1		Tier 2		Tier 3	
A	**Tropical**	f m w s	**Rainforest** **Monsoon** **Savanna, dry winter** **Savanna, dry summer**		
B	**Dry**	W S	**Arid desert** **Semi-arid steppe**	h k	**Hot** **Cold**
C	**Temperate**	w f s	**Dry winter)** **No dry season** **Dry summer**	a b c	**Hot summer** **Warm summer** **Cold summer)**
D	**Continental)**	w f s	**Dry winter)** **No dry season** **Dry summer**	a b c d	**Hot summer** **Warm summer** **Cold summer** **Very cold winter**
E	**Polar**	T F	**Tundra** **Ice cap**		

Calculated based only on the weather patterns of the Core Land (e.g., Denmark is Temperate C-tier, completely ignoring Greenland's E-tier Polar climate).

climate_t3_dominant

What climate Tier 3 is dominant by percentage (according to Tier 3) - Köppen climate classification Scientific Source: Köppen-Geiger Climate Classification System (Peel, Finlayson, and McMahon, 2007).

Tier 1		Tier 2		Tier 3	
A	**Tropical**	f m w s	**Rainforest** **Monsoon** **Savanna, dry winter** **Savanna, dry summer**		
B	**Dry**	W S	**Arid desert** **Semi-arid steppe**	h k	**Hot** **Cold**
C	**Temperate**	w f s	**Dry winter)** **No dry season** **Dry summer**	a b c	**Hot summer** **Warm summer** **Cold summer)**
D	**Continental)**	w f s	**Dry winter)** **No dry season** **Dry summer**	a b c d	**Hot summer** **Warm summer** **Cold summer** **Very cold winter**
E	**Polar**	T F	**Tundra** **Ice cap**		

If there is no Tier 3 classification in the table, write Tier 2 classification.

Calculated based only on the weather patterns of the Core Land (e.g., Denmark is Temperate C-tier, completely ignoring Greenland's E-tier Polar climate).

count_boundaries

Each country's border counts as one. However, if there is a clear separation between two types of borders, they should be counted separately (as two). In the end, the total number of all border types (coastal, inland water, mountain, desert, etc.) must equal this total.

Rule of Border Precedence: It is common for a single boundary to meet the criteria of multiple border types (e.g., a desert border that is also below 1,350m, or a mountain range with a massive river). To prevent double-counting and ensure the total number of specific borders perfectly matches the how_many_boundaries total, coders MUST apply the following strict Order of Precedence. If a boundary fits multiple categories, classify it using the highest applicable rank:

Rank 1: Water Borders (how_many_water_borders / how_many_sea_borders) - Any border defined by a coastline or an inland water barrier (river/lake) wider than 50 meters supersedes all other terrain types.
Rank 2: Mountain Borders (how_many_mountain_borders) - Any land border resting at an altitude above 1,350 meters.
Rank 3: Desert Borders (how_many_desert_borders) - Any border crossing an arid/desert environment (Köppen BWh/BWk).
Rank 4: Loose/Open Borders (how_many_loose_borders) - The default classification for any land border below 1,350 meters that does not feature a 50m+ water barrier or desert terrain.

Warning: Only borders touching the Core Land are counted.

longest_borders

Which border is the longest: sea/water/mountain/dessert/open

border_ratio

Definition: The ratio of the which_border_longest length to the total physical perimeter of the Core Land.
Total Perimeter Rule: Includes all land (open, mountain, water) and sea segments.
Interpretation: * High (> 0.65): Single-terrain dominance (e.g., Italy: 0.79, maritime focus).

- Moderate (0.50–0.65): Transitional geography (e.g., Poland: 0.62, open plain focus).
- Low (< 0.50): High complexity/diverse defenses (e.g., Germany: 0.44).

country_borders

The number of countries that share a border with a given country.

desert_borders

The number of borders that run through desert terrain.

mountain_borders

The number of borders that run through mountainous terrain (above 1,350 meters).

open_borders

Open borders are classified as land borders below 1350 m, with terrain that is easy to cross and no water barrier wider than 50 m.

sea_borders

The number of direct coastal borders.

water_borders

Inland water. The number of borders with rivers or lakes wider than 50 m.

if_independent

- Yes: Sovereign. Full legal and practical independence. Total control over borders, foreign policy, and internal law.
- No (Colony/Occupied): Direct administration by a foreign power. Local governance is either non-existent or completely superseded by the Parent state's military or appointed governors.
- No (Vassal/Tributary): The entity maintains internal local rule but owes formal fealty, military service, or financial tribute to a superior Hegemon/Empire in exchange for protection or the right to exist.
- No (Protectorate): An entity that has surrendered its foreign policy and defense to another state (the Protector) via treaty, while usually retaining domestic administrative autonomy.

system_type

- **Federal:** Power is constitutionally shared between a national government and highly autonomous regional states/provinces (e.g., Modern USA, Modern Germany).
- **Unitary:** Power resides in a central national government. Regional administrations exist, but they act merely as extensions of the state with delegated—not sovereign—power (e.g., Modern France, Modern UK).
- **Central/Command:** Absolute spatial concentration of administration. Regional structures are functionally hollow shells with zero autonomy, existing purely to execute the central capital's orders (e.g., Stalinist USSR, Modern North Korea).

sub_system_type

- **Democratic:** The populace directly decides the foundational direction through highly competitive, multi-party elections. The rule of law supersedes individual leaders (e.g., Modern Nordic states, Roman Republic).
- **Restricted (Transitional/Oligarchic):** Power is heavily concentrated in a specific class, merchant group, or dominant political faction. The playing field is heavily skewed, but civil opposition legally exists and the populace retains the ability to protest or "make demos" (e.g., 18th-century Great Britain, Modern Hungary, Venetian Republic).
- **Absolute:** Total and complete control of the system by a single person, family, or monolithic party. There is zero legal objection allowed, and any opposition is physically liquidated or entirely suppressed (e.g., Ancient Egypt, Modern Belarus, Vatican City).

economic_type

(Distribution of Economic Wealth)

- **Agrarian / Traditional:** Wealth is fundamentally tied to raw land ownership, domestic agricultural output, and human/animal labor. *(Examples: Feudal England, Ancient Rome, Tokugawa Shogunate).*
- **Mercantile / Trade-Centric:** Wealth is generated primarily by controlling external trade routes, taxing commerce, or managing maritime monopolies rather than just farming domestic land. *(Examples: The Dutch Republic, Venetian Empire, Carthage).*
- **Capitalist / Free Market:** Wealth generation and distribution are driven overwhelmingly by private corporate ownership, industrialized free markets, and minimal state intervention. *(Examples: 19th-century Gilded Age USA, Victorian Britain).*
- **Social Capitalism / Mixed Economy:** Wealth is generated by a capitalist free market, but a significant portion is captured and heavily redistributed by the state to ensure social welfare and baseline equality. *(Examples: Modern Nordic Countries, Post-WWII West Germany).*
- **Socialist / Command Economy:** The central government entirely owns, directs, or heavily confiscates the means of production and distribution. Private market generation is illegal or fundamentally suppressed. *(Examples: Soviet Union, Maoist China, Ancient Palace Economies like the Incan Empire).*

if_rich

(Relative Global Prosperity)

Definition: Evaluates the overall economic prosperity, capital accumulation, and general standard of living of the state relative to the rest of the world at that specific time in history. This is strictly comparative. The evaluation must ignore modern standards of living and compare the state only to its global contemporaries in that exact year.

Scientific Baseline (Cliometric Proxies): To prevent qualitative assumptions, this metric MUST be evaluated using established academic frameworks and quantitative proxies where available. When classifying a historical era, base the score on:

1. **Historical GDP Estimates:** (e.g., The Maddison Project Database) for comparative per capita wealth.
2. **Urbanization Rates:** The primary proxy for pre-industrial agricultural surplus (higher urbanization = higher surplus wealth).
3. **Real Wages & Subsistence Ratios:** (e.g., Robert C. Allen's "Bare Bones Basket") measuring how many basic survival baskets an unskilled laborer's wage could purchase compared to other nations.
4. **Energy Capture Index:** (e.g., Ian Morris's Social Development Index) measuring kilocalories of energy captured per capita per day.

Classification Scale (0 to 3):

- **0 - Impoverished / Below Average:** The nation's general prosperity is significantly below the global median of its era. The vast majority of the population lives at basic subsistence levels, struggling with structural poverty, and the state lacks significant surplus capital. *(Examples: Modern developing states in the Global South, 19th-century Qing China during its economic collapse).*
- **1 - Average / Moderate:** The nation aligns with the global median of prosperity for its era. It generates enough surplus to maintain basic stability and a modest standard of living, but it is not a primary center of global or regional wealth. *(Examples: Modern Brazil or Mexico, 18th-century agrarian Eastern European states).*
- **2 - Prosperous / Above Average:** The nation is tangibly wealthier than most of its contemporaries. It possesses a high standard of living, significant capital surplus, robust trade networks, and advanced

infrastructure for its time. *(Examples: Modern South Korea, 16th-century Ming China, 15th-century Florence).*

- **3 - Elite / Vanguard:** The nation sits in the absolute highest percentile of global wealth. It defines the era's economic ceiling, representing the pinnacle of prosperity, capital accumulation, and living standards for that specific historical period. *(Examples: Modern Switzerland or USA, 19th-century Victorian Britain, Golden Age Abbasid Caliphate).*